1. Which mathematical computing and block diagram environment by MathWorks is universally used in electrical engineering education?

(A) MATLAB and Simulink
(B) AutoCAD
(C) Blender
(D) Adobe Illustrator

2. Which specialized Simulink library provides physical modeling blocks for power systems, electrical machines, and power electronics?

(A) Simscape Electrical (formerly SimPowerSystems)
(B) SimEvents
(C) Stateflow only
(D) Simulink Coder

3. What crucial block must be included in any Simscape Electrical Simulink diagram to configure simulation solver methods (Continuous, Discrete, or Phasor)?

(A) powergui block
(B) Scope block
(C) Ground block
(D) Display block

4. Which simulation software suite is the global standard for detailed Electromagnetic Transient (EMT) analysis in power grids?

(A) PSCAD / EMTDC
(B) Microsoft Excel
(C) SolidWorks
(D) ANSYS Fluent

5. Which enterprise software suite is widely used in power utilities for load flow, short-circuit, protection coordination, and arc flash analysis?

(A) ETAP (Electrical Transient Analyzer Program)
(B) Eclipse IDE
(C) Unity Engine
(D) MATLAB Compiler

6. Which electromagnetic finite element analysis (FEA) software by ANSYS is used to design electric motors, transformers, and actuators?

(A) ANSYS Maxwell
(B) ANSYS Mechanical
(C) ANSYS Fluent
(D) ANSYS SpaceClaim

7. Which high-speed simulation software specialized for power electronic circuits uses piecewise linear state-space models?

(A) PLECS
(B) LabVIEW
(C) CorelDRAW
(D) Photoshop

8. Which free, highly popular SPICE simulation software developed by Linear Technology (Analog Devices) is standard in analog circuit design?

(A) LTspice
(B) AutoCAD Electrical
(C) Altium Viewer
(D) Proteus Free

9. What professional electronic design automation (EDA) software suite is widely used for industrial PCB schematic and multi-layer layout?

(A) Altium Designer
(B) Microsoft Paint
(C) LibreOffice Draw
(D) Notepad++

10. Which open-source, community-driven cross-platform EDA software suite is widely adopted for PCB design in university labs?

(A) KiCad
(B) Eagle Freeware
(C) Tinkercad
(D) Fritzing

11. What does PCB stand for in electronics hardware engineering?

(A) Printed Circuit Board
(B) Primary Current Busbar
(C) Power Capacitor Bank
(D) Phase Control Board

12. What standard file format is exported from PCB CAD tools and submitted to fabrication houses for board manufacturing?

(A) Gerber Files (RS-274X)
(B) JPEG Image
(C) MP3 Audio
(D) DOCX Document

13. What parameter in power PCB design defines the shortest path between two conductive parts measured along the surface of the insulation?

(A) Creepage Distance
(B) Clearance Distance
(C) Trace Width
(D) Via Aspect Ratio

14. What parameter defines the shortest distance through air between two conductive parts on a circuit board?

(A) Clearance Distance
(B) Creepage Distance
(C) Drill Hole Diameter
(D) Copper Weight

15. What international safety standard governs insulation coordination and clearance/creepage distances in low-voltage equipment?

(A) IEC 60664
(B) ISO 9001
(C) IEEE 802.3
(D) IP67

16. What test instrument measures electrical voltage, current, and resistance in a single handheld unit?

(A) Digital Multimeter (DMM)
(B) Signal Generator
(C) Spectrum Analyzer
(D) Logic Analyzer

17. What primary test bench instrument displays voltage waveforms as a function of time?

(A) Digital Storage Oscilloscope (DSO)
(B) Frequency Counter
(C) Function Generator
(D) Megohmmeter

18. What specialized probe is required on an oscilloscope to safely measure voltages floating above ground potential in a power electronic converter?

(A) High-Voltage Differential Probe
(B) Standard 1X Passive Probe
(C) BNC Coaxial Cable
(D) Grounding Alligator Clip

19. What instrument measures the high insulation resistance (Megaohms/Gigaohms) of motor windings and power cables?

(A) Megger (Megohmmeter / Insulation Tester)
(B) Digital Multimeter on ohms range
(C) Galvanometer
(D) LCR Meter

20. What instrument measures the loop impedance and resistance of a substation earthing grid?

(A) Earth Resistance Tester (Earth Ground Clamp/Tester)
(B) Voltmeter
(C) Ammeter
(D) Oscilloscope

21. What does GATE stand for in Indian engineering qualifications?

(A) Graduate Aptitude Test in Engineering
(B) General Assessment of Technical Education
(C) Graduate Admission Test for Engineers
(D) Global Assessment of Technology and Engineering

22. Which national test score is used by Indian PSUs (BHEL, NTPC, PGCIL, IOCL) for direct recruitment of graduate engineers?

(A) GATE Score
(B) CAT Score
(C) GRE Score
(D) GMAT Score

23. What Indian premier engineering institutes offer M.Tech and Ph.D. admissions based on GATE percentiles?

(A) IITs (Indian Institutes of Technology) and IISc Bangalore
(B) State Open Universities
(C) Primary Schools
(D) ITI Centers

24. What professional certification is granted by the Bureau of Energy Efficiency (BEE) under the Ministry of Power, India?

(A) Certified Energy Manager (CEM) and Certified Energy Auditor (CEA)
(B) Chartered Accountant
(C) Licensed Electrician Grade 3
(D) Safety Supervisor

25. What globally recognized project management credential is administered by the Project Management Institute (PMI)?

(A) Project Management Professional (PMP)
(B) Six Sigma White Belt
(C) Scrum Master Beginner
(D) ITIL Foundation

26. What premier North American certification board certifies solar photovoltaic installation professionals?

(A) NABCEP (North American Board of Certified Energy Practitioners)
(B) IEEE Standards Board
(C) UL Listed
(D) ANSI Council

27. Which global student engineering competition challenges universities to design, build, and race a formula-style electric racecar?

(A) Formula Student Electric / Formula Bharat EV
(B) World Solar Challenge
(C) Shell Eco-marathon
(D) BAJA SAE Gas

28. In Formula Student Electric, what is the maximum allowable DC tractive system voltage under SAE rules?

(A) 600 V DC
(B) 12 V DC
(C) 1000 V DC
(D) 48 V DC

29. In electric racecar design, what safety circuit immediately shuts down the high-voltage battery contactors upon fault detection?

(A) Tractive System Active Light (TSAL) and Safety Interlock Loop / Shutdown Circuit
(B) Headlight Switch
(C) Stereo Volume Knob
(D) Air Conditioner Clutch

30. What international student competition sponsored by the U.S. Department of Energy challenges teams to design zero-energy solar homes?

(A) Solar Decathlon
(B) Solar Grand Prix
(C) Formula E
(D) NASA Rover Challenge

31. What is India's premier nationwide innovation competition organized by the Ministry of Education for solving real-world government problems?

(A) Smart India Hackathon (SIH)
(B) Techfest
(C) National Science Olympiad
(D) Codeforces

32. Which global student competition challenges teams to achieve the ultimate energy efficiency (highest distance traveled per kWh or liter of fuel)?

(A) Shell Eco-marathon
(B) Formula 1
(C) Indy 500
(D) Baja 1000

33. Which student design challenge organized by the IEEE Power & Energy Society rewards innovative microgrid and renewable energy designs?

(A) IEEE PES Student Design Contest
(B) ACM ICPC
(C) Google Summer of Code
(D) Kaggle Grandmaster Challenge

34. What is the primary function of a Hardware-in-the-Loop (HIL) simulator like Typhoon HIL or OPAL-RT in power electronics R&D?

(A) Emulates physical converters and power grids in real-time on FPGAs to test physical embedded controllers safely without high-voltage hazards
(B) Compiles C++ code to PDF
(C) Draws circuit schematics
(D) Prints 3D plastic enclosures

35. What standard communication bus is used in laboratory bench test instruments to connect to a PC via GPIB, USB, or Ethernet?

(A) VISA (Virtual Instrument Software Architecture) / SCPI
(B) Bluetooth LE
(C) Zigbee
(D) MIDI

36. What Python library provides complete instrumentation control over VISA test equipment?

(A) PyVISA
(B) NumPy
(C) Pandas
(D) Matplotlib

37. What standard human-readable command string queries the instrument identity of any SCPI-compliant multimeter or oscilloscope?

(A) `*IDN?`
(B) `WHO_ARE_YOU?`
(C) `PING`
(D) `GET_MODEL`

38. What does ANSI stand for in electrical engineering standards?

(A) American National Standards Institute
(B) All National Standards Institute
(C) Associated Network Safety Institute
(D) Advanced Numerical System Index

39. What does IEC stand for in international electrical standards?

(A) International Electrotechnical Commission
(B) Indian Electrical Council
(C) Interstate Energy Corporation
(D) Institute of Electronic Controls

40. What Indian national standards body formulates Indian Standards (IS) for electrical equipment and installations?

(A) BIS (Bureau of Indian Standards)
(B) BEE
(C) CEA
(D) CERC

41. In MATLAB/Simulink Simscape Electrical, what is the fundamental difference between 'Continuous' and 'Discrete' solver modes in the powergui block?

(A) Continuous mode solves differential equations with variable time steps; Discrete mode executes fixed-step Tustin/Euler integration for fast, deterministic simulation
(B) Continuous mode is for DC only
(C) Discrete mode cannot model switches
(D) There is no difference

42. What is the 'Phasor' simulation mode in Simscape Electrical powergui designed for?

(A) Simulating electromechanical transient stability and power flow in large multi-machine grids without calculating microsecond switching details
(B) High-frequency RF antenna radiation
(C) Thermal heat dissipation
(D) Single-cell battery chemistry

43. In PSCAD/EMTDC simulation, how are transmission lines and cables modeled to capture frequency-dependent wave propagation accurately?

(A) Frequency-Dependent Phase Model (Bergeron and J.Marti models)
(B) Simple lump resistor
(C) Pure ideal capacitor
(D) Constant inductors

44. What numerical integration method does EMTDC use to solve differential equations of electromagnetic networks?

(A) Dommel's method using the Trapezoidal Rule to convert R, L, C elements into algebraic companion circuits
(B) Forward Euler only
(C) Runge-Kutta 2nd order
(D) Monte Carlo sampling

45. In ETAP, what does an 'Arc Flash Analysis' determine for electrical distribution switchgear?

(A) Incident energy levels (cal/cm^2), arc flash boundaries, and required Personal Protective Equipment (PPE) categories according to IEEE 1584
(B) Switchgear paint thickness
(C) Power factor penalties
(D) Cable scrap value

46. What is a 'Star Coordination Plot' in ETAP protection engineering?

(A) A Time-Current Characteristic (TCC) curve plotting relay trip curves, fuse curves, and cable thermal damage curves to verify selectivity
(B) A stellar astronomical chart
(C) A radar diagram
(D) A 3D motor model

47. In finite element analysis of an electric motor with ANSYS Maxwell, what does the non-linear B-H curve represent?

(A) Magnetic saturation characteristics of the stator and rotor electrical steel laminations
(B) Thermal conductivity of copper
(C) Mechanical tensile strength
(D) Dielectric breakdown of varnish

48. What are 'Core Losses' calculated by ANSYS Maxwell in electromagnetic machines composed of?

(A) Hysteresis loss and Eddy current loss (including excess losses) in the iron core
(B) Friction and windage losses
(C) Ohmic copper loss
(D) Contact resistance

49. Why are laminated electrical steel sheets used in transformer cores and motor stators instead of solid steel blocks?

(A) To break up conductive paths, dramatically reducing circulating eddy currents and core heating losses
(B) To reduce mechanical strength
(C) To increase weight
(D) To make manufacturing difficult

50. In high-frequency power electronics PCB design, what is the critical design rule regarding the power loop area (e.g. switch, diode, and DC link capacitor)?

(A) Minimize loop area to reduce parasitic loop inductance (L_{loop}), preventing high-voltage turn-off ringing ($V = L \frac{di}{dt}$) and EMI emissions
(B) Maximize loop area for cooling
(C) Loop area does not matter
(D) Run loop traces through ground pins

51. What is the purpose of 'Thermal Vias' placed under a surface-mount power MOSFET or voltage regulator IC on a PCB?

(A) Conducts heat away from the semiconductor die through copper-plated holes to an internal or bottom ground copper pour plane
(B) Vents hot air bubbles
(C) Improves electrical insulation
(D) Reduces solder usage

52. What is the standard copper thickness designated as '1 oz copper' on a PCB layer?

(A) 35 micrometers (μm) or 1.37 mils
(B) 70 micrometers
(C) 1 millimeter
(D) 10 micrometers

53. What design precaution must be taken when routing high-speed differential signal pairs (such as CANbus or USB) on a multi-layer PCB?

(A) Maintain tightly matched trace lengths (within millimeters) and constant differential characteristic impedance (e.g. $120\,\Omega$ for CAN, $90\,\Omega$ for USB)
(B) Route traces on opposite sides of board
(C) Add random spiral loops
(D) Cross traces perpendicular to each other

54. Why is an unbroken reference ground plane essential underneath high-speed digital and switching power traces on a 4-layer PCB?

(A) Provides a continuous, low-impedance return current path directly beneath the signal trace, minimizing loop inductance and radiated emissions
(B) Ground planes absorb moisture
(C) Ground planes add structural stiffness only
(D) Ground planes increase circuit resistance

55. What is the difference between an oscilloscope's 'Bandwidth' and its 'Sampling Rate'?

(A) Bandwidth is the analog frequency at which input signal amplitude is attenuated by -3 dB; Sampling rate is the number of ADC samples acquired per second (Samples/sec)
(B) Bandwidth is for DC; Sampling rate is for AC
(C) They are identical metrics
(D) Sampling rate controls screen brightness

56. According to the Nyquist-Shannon criterion, what minimum real-time sampling rate is recommended relative to an oscilloscope's analog bandwidth to reconstruct waveforms accurately?

(A) At least 4 to 5 times the analog bandwidth
(B) Exactly 0.5 times bandwidth
(C) Equal to bandwidth
(D) 100 times bandwidth

57. What does an oscilloscope probe's '10X attenuation' setting achieve compared to '1X'?

(A) Increases input impedance to $10\text{ M}\Omega$ and slashes capacitive loading (to ~10 pF), allowing measurement of high-voltage and high-frequency signals
(B) Amplifies signal by 10x
(C) Causes signal distortion
(D) Limits bandwidth to 1 kHz

58. What is an FFT (Fast Fourier Transform) math function on a modern oscilloscope used for?

(A) Displays the frequency spectrum (magnitude in dB vs frequency) of the measured time-domain waveform to analyze harmonic distortion and EMI
(B) Integrates voltage to find charge
(C) Measures DC current
(D) Saves screenshots

59. What test instrument generates customizable waveforms (sine, square, arbitrary, pulse, ramp) with calibrated frequency and amplitude?

(A) Arbitrary Function / Waveform Generator (AFG / AWG)
(B) Spectrum Analyzer
(C) Frequency Counter
(D) Oscilloscope

60. What test instrument measures the power distribution of an electrical signal across a wide range of radio frequencies?

(A) RF Spectrum Analyzer
(B) Voltmeter
(C) Megger
(D) LCR Meter

61. What instrument measures the precise inductance (L), capacitance (C), and resistance (R) of electrical components at varying test frequencies?

(A) Precision LCR Meter
(B) Ohmmeter
(C) Megohmmeter
(D) Multimeter only

62. In a three-phase power quality analyzer, what standard IEC 61000-4-30 Class A compliance ensures?

(A) Highest precision measurement and standardized aggregation intervals (10 cycles / 2 hours) ensuring legally binding power quality reports
(B) Basic consumer safety
(C) Plastic flame retardancy
(D) Battery life index

63. What electrical power quality parameter measures brief reductions in RMS voltage (typically 10% to 90% dip for 0.5 cycles to 1 minute)?

(A) Voltage Sag (Dip)
(B) Voltage Swell
(C) Transient Surge
(D) Harmonic Intermodulation

64. What power quality parameter measures a temporary increase in RMS voltage above 110% of nominal?

(A) Voltage Swell
(B) Voltage Sag
(C) Flicker
(D) Phase Unbalance

65. What causes 'Voltage Flicker' in industrial distribution networks, and how is it quantified?

(A) Rapid, repetitive voltage fluctuations caused by heavy dynamic loads like electric arc furnaces; quantified by Short-Term Flicker Severity (P_{st})
(B) Loose light bulbs
(C) Low battery charge
(D) Direct current leakage

66. What safety classification on multimeters (CAT I, CAT II, CAT III, CAT IV under IEC 61010) designates tools rated for outdoor service entrance and main utility meters?

(A) CAT IV (highest transient impulse withstand capability, up to 8 kV)
(B) CAT I
(C) CAT II
(D) CAT III

67. In industrial motor testing, what is the 'Blocked Rotor Test' of an induction motor equivalent to in transformer testing?

(A) Short-Circuit Test, determining equivalent leakage impedance and full-load copper losses
(B) Open-Circuit Test
(C) Temperature rise test
(D) Insulation flashover test

68. What does the 'No-Load Test' of an induction motor determine?

(A) Core iron loss and friction & windage mechanical losses
(B) Full-load rotor copper loss
(C) Starting torque
(D) Rotor resistance

69. In high-voltage power transformer testing, what is the 'Tan Delta' (Dissipation Factor) test used for?

(A) Measures dielectric power loss factor ($\tan\delta$) of the insulation system to evaluate moisture contamination and degradation aging
(B) Measures winding turns ratio
(C) Checks oil volume
(D) Measures acoustic noise

70. What test verifies the turns ratio and polarity of high-voltage potential transformers and power transformers?

(A) Turns Ratio Test (TTR)
(B) Hi-Pot DC test
(C) Megger test
(D) Dielectric breakdown test

71. What does 'Hi-Pot' (High Potential) testing determine for newly fabricated electrical machinery and cables?

(A) Verifies that the insulation can withstand severe overvoltages without dielectric puncture breakdown
(B) Measures motor efficiency
(C) Tests cooling fans
(D) Checks bolt torque

72. In embedded motor drives, what hardware peripheral converts optical incremental encoder pulses into rotor position angles without CPU overhead?

(A) Quadrature Encoder Interface (QEI) / Enhanced QEP (eQEP) module
(B) General Purpose Timer in capture mode
(C) UART receiver
(D) ADC converter

73. What communication protocol is the global standard for industrial battery energy storage systems (BESS) and inverter control in microgrids?

(A) Modbus TCP / SunSpec Modbus
(B) HTTP/2
(C) Telnet
(D) Apple AirDrop

74. In hardware development, what is 'Smoke Testing'?

(A) Initial power-up of a newly assembled PCB with a current-limited bench power supply to verify there are no dead shorts or catastrophic component fires
(B) Burning plastic cables
(C) Testing fire alarms
(D) Testing exhaust fans

75. What bench equipment provides adjustable, current-limited, low-noise DC voltages for prototyping electronic circuits?

(A) Regulated DC Bench Power Supply
(B) Car Battery
(C) Wall outlet adapter
(D) Piezo igniter

76. In finite element electromagnetic analysis (FEA), how does the 'Transient with Motion' solver in ANSYS Maxwell couple physical physics?

(A) Solves Maxwell's non-linear field equations coupled with mechanical rotor equations of motion ($J \ddot{\theta} + b \dot{\theta} = T_e - T_L$) and circuit equations at every time step
(B) Solves heat transfer only
(C) Solves static DC fields only
(D) Calculates CAD drawings

77. In high-frequency transformer and inductor design, what mathematical effect causes AC current in one conductor to be constricted by magnetic fields from neighboring conductors?

(A) Proximity Effect, which exponentially increases AC winding resistance beyond standard skin effect predictions
(B) Hall effect
(C) Edison effect
(D) Meissner effect

78. What specialized wire made of multiple independently insulated strands woven together is mandatory for high-frequency transformers to defeat skin and proximity effects?

(A) Litz Wire
(B) Solid Copper Rod
(C) Steel Rebar
(D) Iron Wire

79. In Formula Bharat / Formula Student EV design, how is the Accumulator Management System (AMS) mandated to isolate the tractive system during an insulation fault?

(A) An Insulation Monitoring Device (IMD, e.g. Bender A-ISOMETER) continuously measures chassis isolation resistance; if resistance falls below $500\,\Omega/\text{V}$, it opens the safety shutdown loop
(B) It sounds a horn
(C) It flashes a screen
(D) It dials emergency services

80. What is the primary role of a Pre-Charge Circuit in an electric vehicle high-voltage DC bus?

(A) Slowly charges the large inverter DC bus capacitors through a current-limiting resistor before closing main contactors, preventing destructive contactor welding currents
(B) Charges 12V auxiliary battery
(C) Preheats tires
(D) Heats motor windings

81. What circuit element dissipates high-voltage DC bus energy in an electric vehicle within 5 seconds when the shutdown loop trips?

(A) Active / Passive Discharge Circuit
(B) Brake calipers
(C) Fuse box
(D) Radiator

82. In real-time Hardware-in-the-Loop (HIL) simulators, what is the typical execution time step required to simulate 100 kHz SiC MOSFET converter switching accurately?

(A) Sub-microsecond time steps (typically 100 to 500 nanoseconds executed on dedicated Xilinx FPGAs)
(B) 1 millisecond on CPU
(C) 10 seconds
(D) 1 minute

83. What is 'Control Hardware-in-the-Loop' (C-HIL) versus 'Power Hardware-in-the-Loop' (P-HIL)?

(A) C-HIL interfaces simulated low-voltage signals with physical embedded controllers; P-HIL couples a simulated grid with physical power equipment (e.g. 500 kW real inverter) via high-power 4-quadrant amplifiers
(B) C-HIL is for computers; P-HIL is for phones
(C) There is no difference
(D) P-HIL is purely software

84. In ETAP load flow analysis, why does the Newton-Raphson method fail to converge on ill-conditioned distribution networks with high R/X ratios?

(A) The standard decoupling assumptions break down and the Jacobian matrix becomes near-singular; requiring specialized Newton-Raphson with damping or Forward-Backward Sweep algorithms
(B) ETAP has bugs
(C) R/X ratio is always zero
(D) Power flow is impossible

85. In PSCAD/EMTDC, what is the 'Multiple Run' optimization feature used for in transmission engineering?

(A) Executes automated parameter sweeps (e.g. hundreds of fault angles, breaker closing times) to determine statistical overvoltage distributions and surge arrester energy ratings
(B) Runs multiple copies on printer
(C) Installs updates
(D) Backs up files

86. In high-power PCB design for SiC power modules, what is 'Kelvin Source' connection, and why is it critical?

(A) A dedicated secondary terminal connected directly to the MOSFET source die, separating gate drive return current from high-power drain-source switching current to eliminate source inductance feedback
(B) A ground wire
(C) A temperature sensor
(D) A mounting screw

87. What mathematical metric evaluates the severity of Electromagnetic Interference (EMI) emissions in a conducted emissions test (150 kHz to 30 MHz)?

(A) Quasi-Peak and Average detector measurements in $\text{dB}\mu\text{V}$ using a Line Impedance Stabilization Network (LISN)
(B) Total Harmonic Distortion
(C) Power factor
(D) RMS voltage

88. What is the function of a Line Impedance Stabilization Network (LISN) in automotive and industrial EMC qualification testing?

(A) Provides a standardized $50\,\Omega$ RF impedance across the DUT power terminals and isolates utility power noise from the spectrum analyzer
(B) Filters out DC voltage
(C) Converts AC to 12V
(D) Grounds the operator

89. In an industrial SCADA system, what is the 'Historian' database engine designed for?

(A) Time-series database optimized for high-speed continuous logging, lossless compression, and retrieval of millions of sensor telemetry points per second
(B) Stores employee payroll
(C) Draws user interfaces
(D) Sends email alerts

90. What open communication standard under OPC Foundation enables secure, vendor-neutral data exchange from factory floor PLCs to cloud enterprise systems?

(A) OPC UA (Open Platform Communications Unified Architecture)
(B) Modbus RTU
(C) DNP3
(D) RS-232

91. In building energy modeling for the Solar Decathlon, what simulation engine by the U.S. DOE models whole-building energy consumption, HVAC dynamics, and solar gain?

(A) EnergyPlus / OpenStudio
(B) AutoCAD
(C) SolidWorks
(D) MATLAB Compiler

92. What is a 'Net Zero Energy Building' (NZEB) certified in student green architectural competitions?

(A) A high-performance building where the total annual on-site renewable energy generated equals or exceeds the total annual energy consumed by the building
(B) A building without electricity
(C) A house with zero windows
(D) A building disconnected from sewer lines

93. What is the role of an In-Circuit Programmer/Debugger (such as ST-LINK V3 or J-Link) in embedded firmware development?

(A) Flashes compiled binary firmware into microcontroller flash memory and supports hardware breakpoints and register inspection via SWD/JTAG interfaces
(B) Measures battery voltage
(C) Programs PLC ladder logic
(D) Controls motor speed directly

94. In embedded firmware design, what is 'Static Analysis' using tools like MISRA C compliance checkers?

(A) Analyzing source code without executing it to detect undefined behavior, buffer overflows, and violations of safety-critical embedded coding standards
(B) Testing hardware with a meter
(C) Compiling code on Linux
(D) Measuring execution speed

95. What is 'Code Coverage' (e.g. Statement, Branch, MC/DC) in safety-critical avionics (DO-178C) and automotive (ISO 26262) software verification?

(A) The percentage of source code structures executed by the automated test suite; Modified Condition/Decision Coverage (MC/DC) is mandatory for ASIL-D automotive safety
(B) The length of code comments
(C) The number of functions
(D) The size of binary hex file

96. What is the functional safety standard governing electrical, electronic, and programmable electronic safety-related systems in road vehicles?

(A) ISO 26262 (Automotive Functional Safety)
(B) ISO 9001
(C) IEEE 802.11
(D) IEC 61850

97. In ISO 26262, what is an ASIL rating (ASIL A, B, C, D)?

(A) Automotive Safety Integrity Level, where ASIL D represents the highest degree of automotive hazard requiring the most stringent safety design margins
(B) Car engine horsepower class
(C) Battery capacity grade
(D) Emission sticker

98. What does the 'BEE Star Labeling' program signify on Indian industrial electric motors and distribution transformers?

(A) Energy efficiency rating from 1-Star to 5-Star, where 5-Star represents the lowest energy loss and highest operating efficiency (IE3/IE4 class)
(B) Durability against rain
(C) Number of warranty years
(D) Noise level rating

99. In an industrial substation commissioning project, what is the difference between 'Cold Commissioning' and 'Hot Commissioning'?

(A) Cold commissioning tests de-energized equipment (insulation, wiring continuity, interlocks, relay secondary injection); Hot commissioning energizes equipment at full system voltage and load
(B) Testing in winter vs summer
(C) Testing with heated oil
(D) Air conditioning tests

100. What technical philosophy underpins modern engineering project management in high-stakes competitions and real-world industrial deployments?

(A) Systems Engineering: Decomposing complex multidisciplinary requirements into verifiable subsystems with rigorous interface control, simulation validation, and safety verification
(B) Rushing directly to hardware soldering
(C) Skipping computer simulations
(D) Operating without safety circuits

SECTION B: DESCRIPTIVE & SUBJECTIVE QUESTIONS

Q1. Explain the architecture, simulation solver modes (Continuous, Discrete, Phasor), and use cases of MATLAB/Simulink Simscape Electrical.

Q2. Detail the principles, mathematical modeling, and operational applications of PSCAD/EMTDC in transmission system transients.

Q3. Explain the capabilities and engineering methodology of ETAP for Load Flow, Short-Circuit, and Protection Coordination.

Q4. Analyze finite element electromagnetic analysis (FEA) using ANSYS Maxwell for electrical machine design.

Q5. Detail high-power PCB layout design rules for SiC/GaN power electronic converters: parasitics, creepage, and thermal dissipation.

Q6. Explain the Graduate Aptitude Test in Engineering (GATE): Examination structure, syllabus scope, and PSU recruitment dynamics.

Q7. Detail the Bureau of Energy Efficiency (BEE) Certified Energy Auditor (CEA) qualification: Curriculum, industrial impact, and methodologies.

Q8. Describe the engineering design challenge of Formula Student Electric / Formula Bharat EV, focusing on the High-Voltage Accumulator.

Q9. Explain the architecture of Hardware-in-the-Loop (HIL) test benches for testing digital power electronic controllers.

Q10. Discuss student competitions like Solar Decathlon and Smart India Hackathon (SIH) and their role in developing professional engineering skills.

Q11. Explain the measurement principles of Digital Storage Oscilloscopes (DSOs): Acquisition memory depth, trigger modes, and aliasing.

Q12. Describe the working principles and calibration of a High-Voltage Differential Probe for power electronics measurements.

Q13. Explain the design of high-current power busbars and trace sizing on multi-layer PCBs using IPC-2152 standards.

Q14. Detail the principles of automated instrument control using Python PyVISA and standard SCPI commands.

Q15. Explain the methodology of insulation coordination and impulse withstand voltage testing for electrical equipment.

Q16. Detail the safety protocols and equipment required for electrical arc flash protection under NFPA 70E.

Q17. Describe the operation and calibration of power quality analyzers for harmonic measurement according to IEC 61000-4-7.

Q18. Explain the principles of motor current signature analysis (MCSA) for non-invasive induction motor fault detection.

Q19. Describe the function, layout, and operation of a Low-Voltage Switchgear Motor Control Center (MCC).

Q20. Explain the design and operation of an Insulation Monitoring Device (IMD) in ungrounded IT power systems.

Q21. Discuss the role of technical documentation, engineering drawings, and single-line diagrams (SLDs) in power projects.

Q22. Detail the principles of grounding grid resistance measurement using the Fall-of-Potential method.

Q23. Explain the working and applications of Variable Autotransformers (Variacs) in electrical laboratories.

Q24. Explain the architecture of a real-time battery cell emulator for BMS verification.

Q25. Describe the IEEE Std 1547.1 compliance testing procedures for grid-interactive distributed energy resource inverters.

Q26. Detail the principles of design-for-testability (DFT) in embedded electrical hardware.

Q27. Explain the methodology of partial discharge (PD) testing in high-voltage cables and gas-insulated switchgear.

Q28. Explain the engineering design of a regenerative DC electronic load for testing power supplies and batteries.

Q29. Describe the function and calibration of secondary injection test sets for protective relays.

Q30. Discuss the role of multidisciplinary engineering teamwork in large-scale electrical infrastructure projects.

SECTION C: ANSWER KEY - MULTIPLE CHOICE QUESTIONS

Q1:	[A]	**Q21:**	[A]	**Q41:**	[A]	**Q61:**	[A]	**Q81:**	[A]
Q2:	[A]	**Q22:**	[A]	**Q42:**	[A]	**Q62:**	[A]	**Q82:**	[A]
Q3:	[A]	**Q23:**	[A]	**Q43:**	[A]	**Q63:**	[A]	**Q83:**	[A]
Q4:	[A]	**Q24:**	[A]	**Q44:**	[A]	**Q64:**	[A]	**Q84:**	[A]
Q5:	[A]	**Q25:**	[A]	**Q45:**	[A]	**Q65:**	[A]	**Q85:**	[A]
Q6:	[A]	**Q26:**	[A]	**Q46:**	[A]	**Q66:**	[A]	**Q86:**	[A]
Q7:	[A]	**Q27:**	[A]	**Q47:**	[A]	**Q67:**	[A]	**Q87:**	[A]
Q8:	[A]	**Q28:**	[A]	**Q48:**	[A]	**Q68:**	[A]	**Q88:**	[A]
Q9:	[A]	**Q29:**	[A]	**Q49:**	[A]	**Q69:**	[A]	**Q89:**	[A]
Q10:	[A]	**Q30:**	[A]	**Q50:**	[A]	**Q70:**	[A]	**Q90:**	[A]
Q11:	[A]	**Q31:**	[A]	**Q51:**	[A]	**Q71:**	[A]	**Q91:**	[A]
Q12:	[A]	**Q32:**	[A]	**Q52:**	[A]	**Q72:**	[A]	**Q92:**	[A]
Q13:	[A]	**Q33:**	[A]	**Q53:**	[A]	**Q73:**	[A]	**Q93:**	[A]
Q14:	[A]	**Q34:**	[A]	**Q54:**	[A]	**Q74:**	[A]	**Q94:**	[A]
Q15:	[A]	**Q35:**	[A]	**Q55:**	[A]	**Q75:**	[A]	**Q95:**	[A]
Q16:	[A]	**Q36:**	[A]	**Q56:**	[A]	**Q76:**	[A]	**Q96:**	[A]
Q17:	[A]	**Q37:**	[A]	**Q57:**	[A]	**Q77:**	[A]	**Q97:**	[A]
Q18:	[A]	**Q38:**	[A]	**Q58:**	[A]	**Q78:**	[A]	**Q98:**	[A]
Q19:	[A]	**Q39:**	[A]	**Q59:**	[A]	**Q79:**	[A]	**Q99:**	[A]
Q20:	[A]	**Q40:**	[A]	**Q60:**	[A]	**Q80:**	[A]	**Q100:**	[A]

MCQ Scoring Guide & Difficulty Distribution:

Questions 1 to 40: Foundational & Conceptual [Easy] | Questions 41 to 75: Analytical & Applied [Moderate]

Questions 76 to 100: Complex Numerical & Advanced Problem Solving [Difficult]

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current rise: $\frac{di}{dt} > 5\text{ kA/}\mu\text{s}$. Any parasitic inductance in the power loop (L_{loop}) induces massive voltage overshoots: $V_{overshoot} = L_{loop} \frac{di}{dt}$. Layout designers must place high-frequency ceramic decoupling capacitors immediately adjacent to switch terminals, routing forward and return traces on overlapping adjacent layers to achieve magnetic field cancellation and keep $L_{loop} < 5\text{ nH}$.

2. Creepage and Clearance Coordination (IEC 60664-1):
High-voltage rails (400V/800V DC) require strict physical separation along board surfaces (creepage) and through air (clearance) based on pollution degree and Comparative Tracking Index (CTI). Designers mill isolation slots into the PCB beneath optocouplers and transformers.

3. Thermal Dissipation Management:
Surface-mount power devices dissipate heat into the PCB. Dense arrays of thermal vias (0.3 mm drill diameter, spaced 1 mm apart) conduct heat from top copper pads to internal 2 oz ground planes and aluminum baseplates.

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mandates periodic, comprehensive energy audits for all 'Designated Consumers' (heavy energy-intensive industries: thermal plants, cement, fertilizers, steel, railways).

2. Four-Paper Examination Curriculum:
- Paper 1: General Energy Management: Global energy scenarios, energy conservation act, financing, energy economics (NPV, IRR, Payback Period), and project management.
- Paper 2: Energy Efficiency in Thermal Utilities: Boiler combustion efficiency, heat exchangers, steam traps, furnaces, cogeneration, and waste heat recovery.
- Paper 3: Energy Efficiency in Electrical Utilities: Electric motors (IE3/IE4 standards), VFD applications, transformers, compressed air systems, cooling towers, and commercial HVAC/lighting.
- Paper 4: Energy Performance Assessment (Practical Case Studies): Open-book numerical exam requiring full industrial energy audit balance calculations.

3. Industrial Audit Methodology:
Certified auditors measure electrical power profiles using portable power analyzers, identify thermodynamic and electrical losses, calculate specific energy consumption (SEC), and formulate cost-effective Energy Conservation Measures (ECMs).

Testing high-power converter controllers directly on physical 500 kW hardware risks catastrophic equipment explosion, component destruction, and personnel injury during firmware debugging.

2. Closed-Loop HIL Test Bench Structure:
- Physical Device Under Test (DUT): The real physical microcontroller, DSP (e.g. TI C2000), or FPGA running production C firmware.
- Real-Time HIL Simulator (e.g. Typhoon HIL, OPAL-RT): An ultra-fast multi-core FPGA simulator that executes physical circuit equations (converter switches, filters, electric motor, and utility grid) in real-time with sub-microsecond latency ($< 1\,\mu\text{s}$).
- I/O Interface: The physical controller outputs actual digital PWM gate pulses to the HIL simulator. The simulator reads the PWM signals, simulates converter states, and outputs analog signals (simulated voltages, phase currents, resolver angles) back to the controller's ADC pins in real-time, completing a fully closed hardware feedback loop.

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3. Trigger Modes (Auto, Normal, Single):
- Edge, pulse-width, and runt triggering synchronize the horizontal sweep with specific waveform events, capturing intermittent circuit anomalies.

```
# 2. Configure Digital Multimeter via SCPI commands
dmm.write('*RST') # Reset instrument to factory defaults
dmm.write(':CONF:VOLT:DC 10, 0.0001') # 10V range, 0.1mV resolution

# 3. Execute measurement query
voltage_str = dmm.query(':READ?')
voltage = float(voltage_str)
print(f'Measured Voltage: {voltage:.4f} V')
dmm.close()
```

PyVISA wraps operating system VISA libraries, providing vendor-agnostic control over GPIB, USBTMC, and Ethernet test equipment.

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1. Windowing & FFT Synchronization: Power quality meters synchronize sample windows precisely with fundamental power frequency using a phase-locked loop over a standardized 10-cycle (for 50 Hz) or 12-cycle (for 60 Hz) window ($ pprox 200\text{ ms}$).
2. Harmonic Grouping: Groups discrete DFT frequency bins into harmonic subgroups to include interharmonics without spectral leakage errors, calculating Total Harmonic Distortion (THD) and individual harmonic orders up to the 50th harmonic.

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>>Solution Steps in power projects.

>>Solution Stone Questions Bank inverters.

>>**Solution Statement Question 28:** Assess the function and calibration of secondary windings in large-scale electrical infrastructure projects.